Fail-Safe In-Orbit Reprogramming of a Spacecraft Controller

Hazard Analysis

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1.0 Purpose and Scope

This document is the preliminary hazard analysis of the System: its hazards, their causes and severity, and the controls proposed against them.

The hazard analysis is delivered once per safety phase, each closed by a safety review [2, Secs. 3.1 and 4]:

Phase 0 Concept complete: the top hazards and their causes.
Phase 1 Requirements nearly complete: all known hazards, with their causes, controls and risk.
Phase 2 Design mostly complete: the approved controls, and how each is verified.
Phase 3 Testing complete: proof that the verifications passed.

This document is the first delivery: phases 0 and 1, which small projects combine [2, Sec. 4]. The controls are proposals that the requirements turn into rules; each already names its verification method [7, App. F.3, p. 348].

It covers the software items of the Software Classification [6, Sec. 3.0] in operation: all controller modes, during and between contacts. Bench and build tools are covered where they can reach operation. Its hazards decide which software is safety-critical; the Software Classification records that designation [6, Sec. 4.0].

How to read it. Section 4.0 lists every hazard, how severe it is, and whether it is tolerant enough. Section 5.0 explains in plain words why some are not, and what has to change. Section 6.0 gives one report per catastrophic or critical hazard. The appendices hold the analyses behind them: the fault tree, the failure tables, the glossary and the open items.

2.0 Documents

[1] NASA, *Software Assurance and Software Safety Standard*, NASA-STD-8739.8B, 2022. Causes cite App. A, Table 2, pp. 62–68.
[2] NASA, *NASA Software Engineering Handbook*, NASA-HDBK-2203, Version D, Topic 8.58 “Software Safety and Hazard Analysis”.
[3] NASA, *NASA Software Engineering Handbook*, NASA-HDBK-2203, Version D, Topic 8.05 “SW Failure Modes and Effects Analysis”.
[4] NASA, *Fault Tree Handbook with Aerospace Applications*, Version 1.1, August 2002.
[5] *Fail-Safe In-Orbit Reprogramming of a Spacecraft Controller — Concept of Operations*, Issue 1, Revision 1, Draft.

[6] *Fail-Safe In-Orbit Reprogramming of a Spacecraft Controller — Software Classification*, Issue 1, Revision 1, Draft.
[7] NASA, *NASA Software Safety Guidebook*, NASA-GB-8719.13, 2004.
[8] NASA, *NASA Systems Engineering Handbook*, NASA/SP-2016-6105 Rev 2, 2016.
[9] NASA, *Fault Management Handbook*, NASA-HDBK-1002, Draft 2, 2 April 2012. A draft, not a released standard.

3.0 Ground Rules

These rules set the boundary, the failures included, the fault tolerance required and how failures are modelled and named, before the analysis starts [3, Sec. 3.1; 4, pp. 23–24].

3.1 Boundary

- **Harm:** the loss of the controller's operability, observability or correctability.
- **Boundary:** the radio equipment and the spacecraft power supply lie outside the System [5, Sec. 1.2.2]. Their failures enter as causes and are not analysed further.
- **Physical tampering:** excluded; nobody can reach the spacecraft (Assumption A-1).

3.2 Failures Included

The failures below are taken from the software causes of hazards [1, App. A, Table 2, pp. 62–68].

Domain	Failures
Software	Hang, crash, buffer overflow or invalid state in any software item
Data	Message corrupted, lost, repeated or late on any link; full buffer
Memory	Write cut short by power loss; single-event upset (a bit flipped by radiation, A-5); corrupted value that survives a restart
Timing	Watchdog timeout; restart loop; onboard clock wrong, drifting or reset
Power	Unannounced loss at any moment (A-2)
Hardware link	Physical failure of a data-link connection or the reprogramming link
Operator	Invalid, out-of-sequence or mistimed telecommand (A-4)
Security	Means of proving origin disclosed

3.3 Analysis Rules

This section defines the evaluation rules, fault tolerance criteria, and baseline assumptions governing the formal hazard and fault tree analyses [3, Sec. 3.1; 7, Sec. 3.2.1].

- **Counting failures:** to check a hazard, we count the failures that would have to happen for it to occur. If one cause disables several controls at once, it counts as one failure, not one per control. Counting each control separately would make the System look more tolerant than it is.
- **Severity:** each hazard is rated by how bad its outcome is if it happens, using the four categories in the legend below. The rating ignores the controls: they make a hazard less likely, but its outcome is just as bad when it does happen. A hazard therefore keeps its severity however many controls it has.
- **Fault tolerance:** how many failures must happen together before a hazard can occur. A catastrophic hazard must need at least three of the failures in Section 3.2 together, a critical hazard at least two [3, Sec. 3.1]. For each hazard, we find the smallest set of failures that causes it and compare it with this number. A hazard that needs fewer breaks the rule, and its controls must be strengthened.

- **Safety effort:** full [7, Sec. 3.2.1, p. 44], because between contacts no operator can respond to a hazard (A-3). It decides the analyses here: a fault tree is mandatory and a failure modes analysis highly recommended [7, Table 6-3, p. 107].
- **Assumptions:** the analysis relies on the ConOps assumptions A-1 to A-5, and adds two of its own, each taken as the worst case:
  1. the last-resort firmware the restore puts back has the same start-up code, defects and kept data as the other images, so a cause that stops every image also stops the restored one;
  2. one means proves that both commands and firmware come from the operator, and both the onboard computer and the controller use it, so disclosing it opens both.

Legend: severity (categories from [7, Table 2-2, p. 27])

CATASTROPHIC	The controller is lost for the rest of the mission: it cannot run, or can no longer be commanded, observed or corrected.
CRITICAL	The controller acts against the operator's intent, or cannot be commanded, observed or corrected until the operator acts at a later contact.
MODERATE	Part of a function is lost, such as some health reports or one update attempt; the controller still runs and obeys the operator.
NEGLIGIBLE	Nothing is lost; the operator has extra work or gets data a contact later.

3.4 Naming

- **Hazards:** H-1 to H-17, in order of severity.
- **Fault tree:** gates G01 to G06; primary events B01 to B10. No identifier is ever reused.
- **ConOps numbers:** objectives (O), assumptions (A), constraints (C), risks (R), capabilities and scenarios refer to the ConOps [5, Secs. 1.1.2, 3.1, 3.5, 6.0 and 8.0]. A control with a scenario number comes from the ConOps; one without is new.
- **Software role** [2, Sec. 3.2.2]: CAUSE, a software fault can lead to the hazard; CONTROL, software prevents it; MITIGATION, software limits its effect; MONITOR, software detects and reports it.
- **Verification method** [8, Sec. 5.3.1.2.1, p. 92]: TEST, using the product to obtain detailed data; DEMONSTRATION, using it to show the control works, without detailed data; INSPECTION, examining the product or its records; ANALYSIS, modelling or calculation.

4.0 Preliminary Hazard List

The hazards in order of severity **[TBR-1]**. For each catastrophic and critical hazard, the table also gives the fewest failures that cause it, with the proposed controls working, and whether that meets the fault tolerance rule (Section 3.3).

ID	Hazard	Severity	Fewest	Meets rule
H-1	The controller is left with no firmware it can run, and cannot be restored	CATASTROPHIC	1	NO
H-2	Firmware from someone other than the operator runs on the controller	CATASTROPHIC	1	NO
H-3	New firmware can no longer reach the controller	CATASTROPHIC	1	NO
H-4	The controller can no longer be commanded or observed	CATASTROPHIC	1	NO
H-5	Someone other than the operator commands the System	CATASTROPHIC	1	NO

ID	Hazard	Severity	Fewest	Meets rule
H-6	A command is acted on in a form or at a time the operator did not intend	CRITICAL	1	NO
H-7	Wrong firmware runs on the controller: damaged, out of date, or with a hidden defect	CRITICAL	2	YES
H-8	The controller stops working between contacts and is not brought back	CRITICAL	1	NO
H-9	The ground is shown a false state of the System	CRITICAL	1	NO
H-10	A fault between contacts never reaches the ground	CRITICAL	2	YES
H-11	A valid command is not acted on, and the ground is not told	CRITICAL	2	YES
H-12	A test function acts outside a test	CRITICAL	2	YES
H-13	A fault response is taken when there is no fault	MODERATE	N/A	N/A
H-14	An update cannot be completed	MODERATE	N/A	N/A
H-15	Health reports are lost	MODERATE	N/A	N/A
H-16	Reports reach the ground later than planned	NEGLIGIBLE	N/A	N/A
H-17	A command is refused with a wrong or unclear reason	NEGLIGIBLE	N/A	N/A

Legend. Fewest: the fewest failures that together cause the hazard. Meets rule: YES when Fewest is at least three for a catastrophic hazard or two for a critical one; NO otherwise. N/A: the rule does not cover moderate and negligible hazards.

5.0 Findings

Eight of the twelve catastrophic and critical hazards can happen from a single failure, so they do not meet the fault tolerance rule yet. Each time, the cause is shared: one event defeats several controls at once. The analysis looked for such shared causes, including the same defect in software that runs in more than one place [1, App. A.1.1, p. 62]. They are listed below with the hazards they reach; those that break the rule come first, each with what the requirements must change.

5.1 Causes That Break the Rule

- **Shared image code and data.** Every firmware copy, including the last-resort firmware the restore puts back, shares the start-up code, the data kept over restarts and any defect they have in common (assumption 1). One fault there stops them all. Breaks H-1 and H-8; also reaches H-6 and H-13. *To change:* the last-resort firmware and the restore must not depend on what the other copies share.
- **One proof of origin.** One means proves that both commands and firmware come from the operator (assumption 2). If it is disclosed, another party can command the controller and load firmware. Breaks H-2 and H-5. *To change:* a disclosure must be noticed, so the means can be replaced in flight (Scenario-370).
- **Fixed onboard computer software.** It is not updated during the mission [5, Sec. 5.0], so a defect in its update path stays for good. Breaks H-3 and H-4; also reaches H-1. *To change:* a decision for the author: accept this risk, or require that the onboard computer software can be corrected.

- **One onboard computer.** It is the only path for commands, reports and restores; if it fails for good, all three stop (R-4). Breaks H-4; also reaches H-1, H-8 and H-10. *To change:* a decision for the author: accept this risk, or require a second path to the controller.
- **Same data-link software.** The two data-link connections share software at both ends, so one defect cuts both (R-1) [1, p. 66]. Breaks H-4. *To change:* one defect must not be able to cut both connections.
- **Wrong onboard clock.** Expiry, arm time-outs, scheduled commands, time-tags and the trial time limit all depend on it [1, p. 65]. A wrong clock alone runs a scheduled command at the wrong time and gives reports wrong times. Breaks H-6 and H-9; also reaches H-4, H-11 and H-13. *To change:* no check may depend on the onboard clock alone.
- **Radiation.** A bit error can change a held command after it was checked and before it runs (A-5) [1, p. 63]. Breaks H-6; bit errors also reach H-1, H-7, H-8, H-9 and H-10. *To change:* a held command is checked again just before it runs.

5.2 Causes Within the Rule

These defeat several controls at once, but no hazard falls to them alone.

- **One power supply.** It feeds both the onboard computer and the controller [5, Sec. 3.3]; one loss stops the controller and the computer that would restore it. Reaches H-1, H-8, H-10, H-11 and H-15.
- **One set of build tools.** The same tools build every firmware copy; a tool defect enters all of them. Reaches H-1, H-7 and H-12.

6.0 Hazard Reports

Reports cover the catastrophic and critical hazards, the two categories of the hazard report form [7, Fig. F-1, p. 351]; the others stay in the preliminary hazard list. Each report gives the conditions, worst effect, causes, proposed controls with their verification method, and how the controls fail [2, Sec. 4]. The shared causes behind them are in Section 5.0.

H-1 The controller is left with no firmware it can run, and cannot be restored. CATASTROPHIC. Software: CAUSE, CONTROL.
When: An update, a trial, any restart, or a needed restore. **Effect:** The controller is lost for the rest of the mission (A-1).
Causes: (a) power lost while firmware is written (Scenario-600) [1, p. 63]; (b) a bit error in stored firmware (Scenario-610) [1, p. 63]; (c) an update writes over the previous or last-resort firmware [1, p. 65]; (d) new and previous firmware both fail (Scenario-460); (e) the last-resort firmware has changed unnoticed [1, p. 63]; (f) a restart loop never reaches the fall-back [1, p. 64]; (g) the start-up code fails or is damaged [1, p. 63]; (h) the reprogramming link fails, or the onboard computer misses the need to restore (Scenario-520); (i) the onboard computer's known-good copy is damaged [1, p. 63].
Controls: (1) an update never writes over the previous or last-resort firmware (Scenario-600) TEST; (2) every image is checked before it runs (Scenario-610) TEST; (3) after a set number of failed starts, the controller returns to its previous firmware (Scenario-440) TEST; (4) a last-resort firmware that updates never replace (Scenario-460) TEST; (5) a restore of the last-resort firmware through the reprogramming link, independent of the controller's software (Scenario-480) TEST; (6) stored firmware and the reprogramming link are checked between uses TEST; (7) a power loss during the trial counts as a failed start (Scenario-605) TEST; (8) failed firmware is not started again unless the operator orders it (Scenario-430 to Scenario-460) TEST.
Fails if: The image check passes damaged firmware; the failed-start count is lost at a restart; the last-resort firmware, or the image the restore puts back, shares the defect.

H-2 Firmware from someone other than the operator runs on the controller. CATASTROPHIC. Software: CONTROL.
When: An update or a restore. **Effect:** Hostile firmware runs the controller and can refuse updates and report a false state. The restore starts only when no firmware can run (Scenario-480), so this firmware is never removed.
Causes: (a) firmware not from the operator arrives (Scenario-410) [1, p. 67]; (b) the means of proving origin is disclosed; (c) a restore starts a stored copy someone else replaced.
Controls: (1) the onboard computer refuses firmware not released by the operator (Scenario-410) TEST; (2) the controller checks the origin again before it runs the firmware TEST; (3) the operator can order a restore while firmware runs DEMONSTRATION; (4) a disclosed means of proving origin is replaced in flight (Scenario-370) TEST.
Fails if: The check accepts what it should refuse; the means of proof is disclosed without the operator knowing, so it is never replaced.

H-3 New firmware can no longer reach the controller. CATASTROPHIC. Software: CAUSE, CONTROL.
When: After an update is confirmed, or after a fault in the onboard computer. **Effect:** The controller can no longer be corrected for the rest of the mission.
Causes: (a) confirmed firmware runs, but cannot receive firmware; (b) a defect in the onboard computer's update path; (c) a wrong stored minimum version or controller identity refuses all firmware [1, p. 67]; (d) firmware storage on the onboard computer stays full [1, p. 66].
Controls: (1) new firmware runs a trial before it is confirmed (Scenario-430 to Scenario-450) TEST; (2) it is judged good only after it has received firmware itself TEST; (3) the last-resort firmware can always receive firmware TEST; (4) the operator can order a return to the last-resort firmware DEMONSTRATION.
Fails if: The trial skips the update path; the defect is in the onboard computer, out of every control's reach.

H-4 The controller can no longer be commanded or observed. CATASTROPHIC. Software: CAUSE, CONTROL.
When: At any time. **Effect:** The controller runs on its current firmware, but nobody can command or observe it for the rest of the mission.
Causes: (a) both data-link connections fail (Scenario-510); (b) one software defect cuts both (R-1) [1, p. 66]; (c) a controller that never stops sending blocks the data link [1, p. 63]; (d) a wrong onboard clock makes every command expire, including the clock correction [1, p. 65]; (e) the ground loses the means of proving origin; (f) a false fault response marks both connections failed (H-13); (g) a replacement of the means of proving origin goes wrong, so no command is accepted (Scenario-370); (h) the onboard computer fails for good (R-4).
Controls: (1) two separate data-link connections, one enough (O-3, Scenario-500) TEST; (2) a lost connection is reported (Scenario-510) TEST; (3) the reprogramming link stays usable (Scenario-510) TEST; (4) the clock correction does not depend on the onboard clock TEST.
Fails if: One cause cuts both connections. No control yet covers a lost means of proving origin, or the loss of the only onboard computer.

H-5 Someone other than the operator commands the System. CATASTROPHIC. Software: CONTROL.
When: During a contact, or at any time for a source on the spacecraft. **Effect:** Another party can do anything the operator can, and hide it, for the rest of the mission.
Causes: (a) a command not from the operator arrives (Scenario-300) [1, p. 67]; (b) the means of proving origin is disclosed; (c) someone else uses the operator interface, and no second person checks a command (A-4); (d) a command arrives from an unexpected system on the spacecraft [1, p. 62].

Controls: (1) every command is proven to come from the operator; any other is refused and logged (O-2, Scenario-300) TEST; (2) only the operator can use the operator interface DEMONSTRATION; (3) a disclosed means of proving origin is replaced in flight (Scenario-370) TEST.
Fails if: The means of proof is disclosed without the operator knowing, so it is never replaced.

H-6 A command is acted on in a form or at a time the operator did not intend. CRITICAL. Software: CAUSE, CONTROL.
When: While a command is checked, held or run. **Effect:** The controller does what the operator did not intend; the ground learns of it at the next contact.
Causes: A command (a) changed in transfer [1, p. 62]; (b) sent again from an old copy (Scenario-300); (c) run twice after a lost answer (Scenario-350); (d) not allowed in the current mode (Scenario-310) [1, p. 63]; (e) with impossible values (Scenario-320) [1, p. 62]; (f) dangerous, without a valid arm (Scenario-330, Scenario-335) [1, p. 63]; (g) past its expiry (Scenario-340) [1, p. 63]; (h) scheduled, run at the wrong time or again after a restart [1, p. 64]; (i) held, or the arm state, changed by a bit error [1, p. 63]; (j) built wrongly by the ground software or entered wrongly by the operator (A-4) [1, p. 63].
Controls: (1) the onboard computer checks that every command comes from the operator, is new and has not expired, and the controller checks that it is allowed in the current mode and has possible values (Scenario-120) TEST; (2) a dangerous command needs an arm, which expires (Scenario-330, Scenario-335) TEST; (3) a repeated command is answered "already done" (Scenario-350) TEST; (4) the ground software checks every command before it is sent (Scenario-120) TEST; (5) every command and answer is logged (Scenario-120) INSPECTION.
Fails if: The checks use a wrong clock; a held command or the arm state changes after its check; new firmware checks commands wrongly (H-7).

H-7 Wrong firmware runs on the controller: damaged, out of date, or with a hidden defect. CRITICAL. Software: CAUSE, CONTROL.
When: During and after an update, a fall-back or a restore. **Effect:** The controller misbehaves until a further update. If the firmware passes its trial, it becomes the known-good firmware the controller falls back to. If it stops updates, H-3 follows.
Causes: (a) a hidden defect the trial does not exercise; (b) firmware changed after its check, in transfer or in storage [1, p. 63]; (c) firmware older than the minimum version sent again (Scenario-420); (d) the wrong build, or wrong configuration values, released [1, p. 67]; (e) firmware built for another controller (Scenario-360); (f) a defect added by the build tools; (g) code released without full testing [1, p. 67].
Controls: (1) firmware not judged good within the trial time is given up (Scenario-430 to Scenario-450) TEST; (2) the onboard computer refuses firmware that is damaged (Scenario-400), older than the minimum version (Scenario-420) or built for another controller (Scenario-360) TEST; (3) the controller checks firmware again before it runs it (Scenario-610) TEST; (4) the ground software checks every release before it is sent (Scenario-100) TEST; (5) the operator can order a return to the previous firmware DEMONSTRATION.
Fails if: The judgement checks too little, or the trial ends before the defect shows; a check is made before the last transfer.

H-8 The controller stops working between contacts and is not brought back. CRITICAL. Software: CAUSE, CONTROL, MONITOR.
When: In any controller mode. **Effect:** The controller's work stops until the ground acts at the next contact.
Causes: (a) a bit error in running memory (Scenario-620) [1, p. 63]; (b) the firmware hangs, deadlocks or overloads [1, p. 63]; (c) a restart does not resume [1, p. 63]; (d) the restart takes longer than the time to the next failure [1, p. 64]; (e) the controller misbehaves when no command arrives for a long time [1, p. 65]; (f) the onboard computer misses the silence, or its watching program has crashed (Scenario-520, Scenario-530).

Controls: (1) the controller restarts on its own and resumes (Scenario-620) TEST; (2) the onboard computer notices the silence, tries to bring the controller back and records it (Scenario-520) TEST; (3) a crashed onboard computer program restarts and resumes (Scenario-530) TEST; (4) new firmware that hangs is given up (Scenario-430) TEST; (5) the controller keeps working while the whole onboard computer restarts (Scenario-535) TEST.
Fails if: The controller's restart and the onboard computer's watch fail from one fault.

H-9 The ground is shown a false state of the System. CRITICAL. Software: CAUSE, CONTROL.
When: At every contact. **Effect:** The operator decides on false information; a harmful command (H-6) or a missed fault (H-10) follows.
Causes: (a) wrong time-tags [1, p. 64]; (b) old reports shown as current [1, p. 63]; (c) a report changed in storage or in transfer [1, p. 62]; (d) "done" or "already done" answered for a command that did not run; (e) the ground software stores or shows a report wrongly [1, p. 62]; (f) after a fall-back, the ground does not know which firmware runs.
Controls: (1) the onboard computer time-tags every report (Scenario-110) TEST; (2) every report is checked for damage on arrival TEST; (3) the ground software keeps every report received [5, Sec. 3.2] INSPECTION; (4) after a fall-back, the next reports tell the ground which firmware runs (Scenario-430 to Scenario-460) TEST.
Fails if: The onboard clock is wrong; a report is damaged before its check.

H-10 A fault between contacts never reaches the ground. CRITICAL. Software: CAUSE, MONITOR.
When: Between contacts. **Effect:** The ground does not know of the fault; hidden faults pile up, such as a damaged last-resort firmware (H-1).
Causes: (a) the fault is not detected [1, p. 66]; (b) no event report is made [1, p. 65]; (c) the report is lost to full storage, a crash or a power loss (Scenario-210, Scenario-530, Scenario-535); (d) the ground loses it.
Controls: (1) event reports are kept when storage is full (Scenario-210) TEST; (2) no report is lost without the ground being told (O-5) TEST; (3) stored reports survive a power loss and a restart, including a restart of the whole onboard computer (Scenario-535) TEST.
Fails if: Fault detection depends on the part that has failed.

H-11 A valid command is not acted on, and the ground is not told. CRITICAL. Software: CAUSE, CONTROL.
When: While a command is checked, held or run. **Effect:** The operator believes an action was taken that was not, for example a cancelled update (Scenario-470).
Causes: (a) a full buffer drops the command [1, p. 63]; (b) a wrong onboard clock expires it (Scenario-340); (c) a scheduled command is lost at a restart (Scenario-130); (d) data-link traffic blocks it [1, p. 63]; (e) its answer is lost (Scenario-350).
Controls: (1) every command gets two answers, received and then done with the result (Scenario-120) TEST; (2) a refusal carries its reason (Scenario-310 to Scenario-340) TEST; (3) scheduled commands survive a restart TEST.
Fails if: The answers are lost with the command.

H-12 A test function acts outside a test. CRITICAL. Software: CAUSE.
When: In operation. **Effect:** Power is cut, a data-link connection is broken or the radio link is degraded in operation; each is a cause of H-1 or H-4.
Causes: (a) fault-injection functions stay in the operating software [1, p. 64]; (b) a wrong command, or one not from the operator, starts them.
Controls: (1) test functions serve verification only and are not part of the operating software (Capability-5) INSPECTION.
Fails if: A bench build is released for operation [1, p. 67].

7.0 Must-Work and Must-Not-Work Functions

The hazard analysis identifies the functions that must work and those that must not work [1, App. A.1.2, p. 62]. Both lists hold in operation.

Must work: return to working firmware when new firmware fails (H-1); restore through the reprogramming link (H-1, H-2); check every image before it runs (H-1, H-2, H-7); refuse commands and firmware not from the operator (H-2, H-5); accept the clock correction (H-4); detect and report faults between contacts (H-8, H-10); keep and send down every event report (H-10); answer every command (H-11); replace a disclosed means of proving origin (H-2, H-5).

Must not work: write over the previous or last-resort firmware (H-1); run firmware not from the operator (H-2); run a command not from the operator (H-5); run a dangerous command without a valid arm, or any command twice (H-6); confirm firmware not judged good (H-7); inject faults (H-12); accept a new means of proving origin not proven with the current one (H-4, H-5).

A Top-Down Analysis

Figure 1 develops the worst hazard, H-1. Each box states what fails and when [4, p. 50].

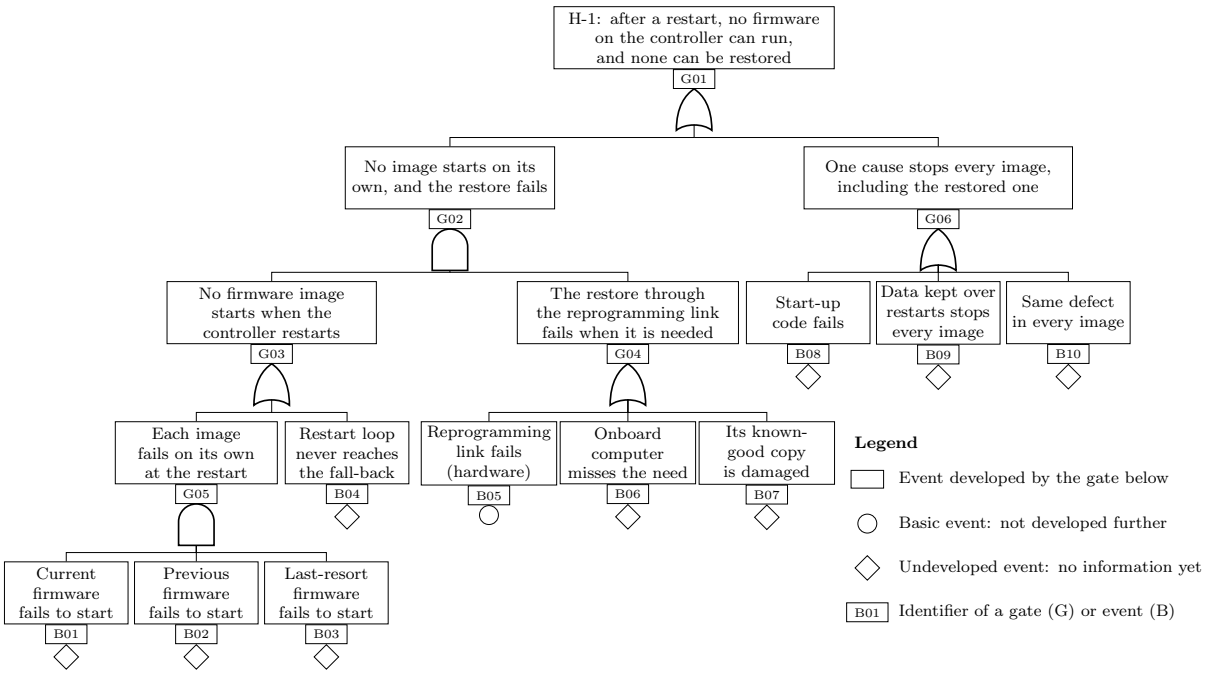


Figure 1: Fault tree of H-1. Symbols as in [4, Table 4-1, p. 34] and [4, Fig. 4-1, p. 33].

H-1 has three kinds of minimal cut set: one event (B08, B09 or B10); two events (B04 with one of B05 to B07); four events (B01, B02 and B03 with one of B05 to B07). By assumption 1 (Section 3.3), B08, B09 and B10 also stop the image the restore puts back, so each causes H-1 alone.

Section 4.0 gives the same count for every catastrophic and critical hazard.

B Bottom-Up Analysis

Before a design exists, the components examined are the System's functions [3, Sec. 4]. Failures are listed in an events table and a data table [3, Sec. 5.3]: each function for each way it can fail,

and each item of data for each way it can be bad. Each failure is traced to the hazard it leads to.

Function	HALT	OMISSION	INCORRECT	TIMING	UNWANTED
Telemetry (Capability-1)					
Make health and event reports	H-10, H-15	H-10, H-15	H-9	H-9	H-4, H-15
Time-tag and store reports	H-10, H-15	H-10, H-15	H-9	H-9	H-15
Send reports down	H-16	H-10, H-15	H-9	H-16	H-16
Keep received reports on the ground	H-10, H-15	H-10, H-15	H-9	NONE	NONE
Command handling (Capability-2)					
Prepare and check commands on the ground	H-11	H-6	H-6	H-6	H-6
Check commands onboard	H-11	H-5, H-6	H-6, H-11, H-17	H-4, H-6, H-11	NONE
Hold armed and scheduled commands	H-11	H-11	H-6	H-6	H-6
Run commands	H-8, H-11	H-11	H-6	H-6	H-5, H-6
Replace the means of proving origin	H-4	H-2, H-5	H-4	H-4	H-4, H-5
Answer and log commands	H-9	H-11	H-9, H-17	H-16	H-9
Fault handling (Capability-3)					
Detect faults	H-8, H-10	H-8, H-10	H-10, H-13	H-8	H-13
Respond to faults	H-1, H-8	H-8	H-1, H-4, H-13	H-8	H-13
Update (Capability-4)					
Prepare and check firmware releases	NONE	H-7	H-3, H-7	NONE	H-7
Receive and check firmware	H-14	H-1, H-2, H-7	H-2, H-3, H-7, H-14	H-7	H-14
Install firmware and switch	H-1	H-14	H-1	H-1	H-6
Judge and confirm new firmware	H-13	H-13	H-3, H-7	H-7, H-13	H-7
Fall back to working firmware	H-1, H-8	H-1, H-8	H-1, H-7	H-8, H-13	H-13
Restore through the reprogramming link	H-1	H-1	H-1, H-2, H-7	H-1	H-13
Test (Capability-5)					
Inject faults	NONE	NONE	NONE	H-12	H-12

Legend: event failure type (types from [7, App. D.4.8, p. 336]). HALT: the function stops abnormally. OMISSION: the function does not happen. INCORRECT: it happens with wrong logic or a wrong result. TIMING: it happens at the wrong time or out of sequence. UNWANTED: it happens when it is not expected. NONE: leads to no hazard.

Data	OUT OF RANGE	MISSING	OUT OF SEQUENCE	OVERWRITTEN	WRONG
Command	H-6	H-11	H-6	H-6	H-6
Arm state	H-6	NONE	NONE	H-6	H-6
Firmware image	H-7	H-14	H-7, H-14	H-1, H-7	H-2, H-7
Report	H-9	H-10, H-15	H-9	H-10, H-15	H-9
Onboard time	H-4, H-6, H-9	H-6, H-9, H-11	H-6, H-9	H-6, H-9	H-4, H-6, H-9, H-11, H-13
Kept data	H-1, H-3	H-1, H-13	NONE	H-1, H-3, H-13	H-1, H-3, H-13

Data	OUT OF RANGE	MISSING	OUT OF SEQUENCE	OVERWRITTEN	WRONG
Means of proving origin	NONE	H-4	NONE	H-4	H-4

Legend: data failure type (types from [7, App. D.4.8, p. 335]). OUT OF RANGE: a value outside its possible range. MISSING: the data is absent. OUT OF SEQUENCE: the data arrives or is used in the wrong order. OVERWRITTEN: valid data is replaced. WRONG: the data is valid in form but incorrect. NONE: leads to no hazard. Kept data: the counts, flags, minimum version and controller identity kept over restarts.

The effect of each failure is the hazard in its cell; its severity, detection and compensation are those of that hazard. Every failure leads to a hazard of Section 4.0, or to none.

C Glossary

Common cause A cause that defeats several controls at once; it can negate a redundancy [4, Sec. 1.3, p. 3].

Control Hazard control: "means of reducing the risk of exposure to a hazard" [1, p. 13].

Fault tolerance "The ability to perform a function in the presence of any of a specified number of coincident, independent failure causes of specified types" [9, pp. 18–19].

Hazard "A state or a set of conditions, internal or external to a system that has the potential to cause harm" [1, p. 13].

Minimal cut set "The smallest combination of basic events that result in the top event" [4, Sec. 1.3, p. 3].

D Open Items

The values below are not yet known.

TBD-1 Project name and document reference.

The author accepted the severity rule, the meaning of each severity category and the full safety effort on 10 October 2026, and the fault tolerance rule and the two assumptions of this analysis on 11 October 2026 (Section 3.3). The proposals below await the author's acceptance.

TBR-1 The hazard list and the severity of each hazard.